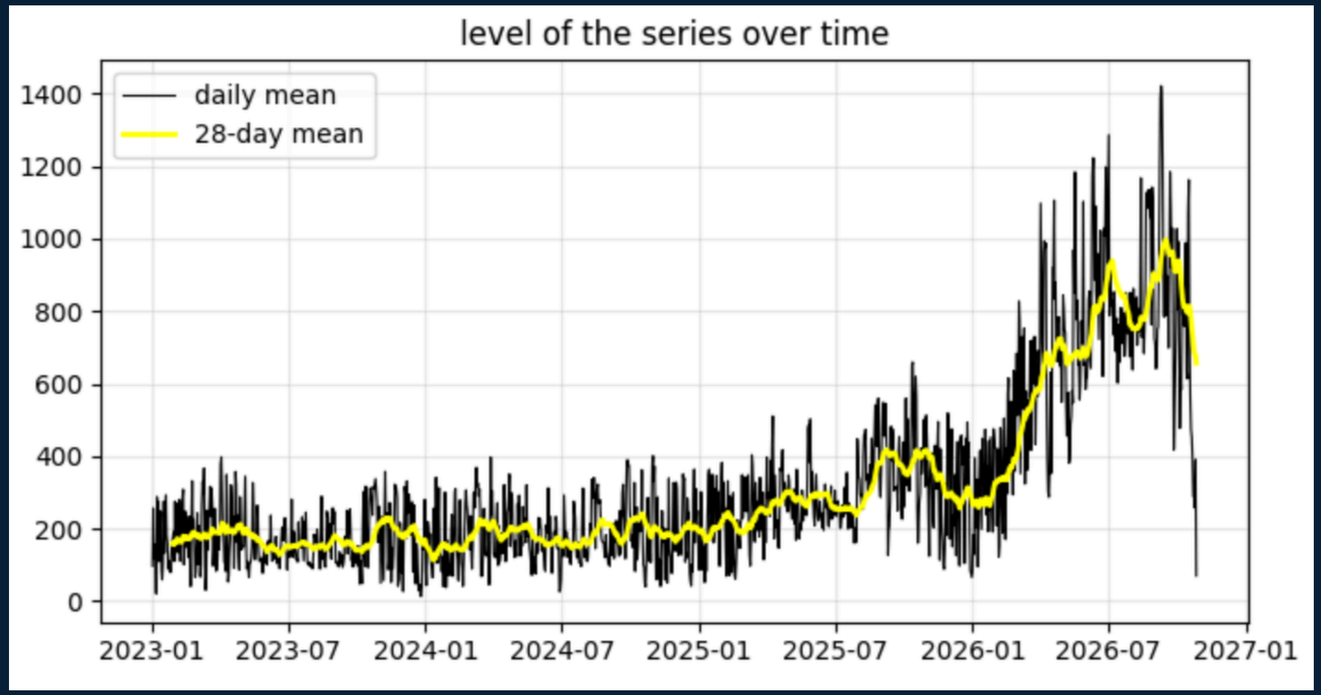
48 hours of generation

Mariana Konovalenko

Our data

The level of this time series grew from ~150 (2023) to ~900 (summer 2026), then dropped to ~300 in late October.
Main points of the dataset:

- Night generation is driven by wind: ~150 at 0 m/s, ~440 at 5 m/s
- daily mean autocorrelation 0.89, so seasonal naive is hard to beat



First results

THE SESSION 1 MODEL

LightGBM + 25% seasonal naive, interval from quantile LightGBM with CQR

PUBLIC LEADERBOARD: 29TH

On the first hidden day it lost on all three metrics. RMSE 122 vs 116 and MAPE 87% vs 84% were close, but Winkler was 917 vs 521

VALIDATION LOOKED GOOD

On 10 validation windows the model beat seasonal naive: RMSE 228 vs 267, and the interval was much better than the benchmark's

WHY IT LOST

The interval was calibrated when the level was much higher, so it was far too wide. One calm day showed it, but a single day cannot decide which model is better

Final Model

POINT FORECAST

Two LightGBM models, averaged: one predicts y, the other y divided by its mean over the last 7 days, so the model follows the changing level. Day 1 and day 2 have separate models

INTERVAL

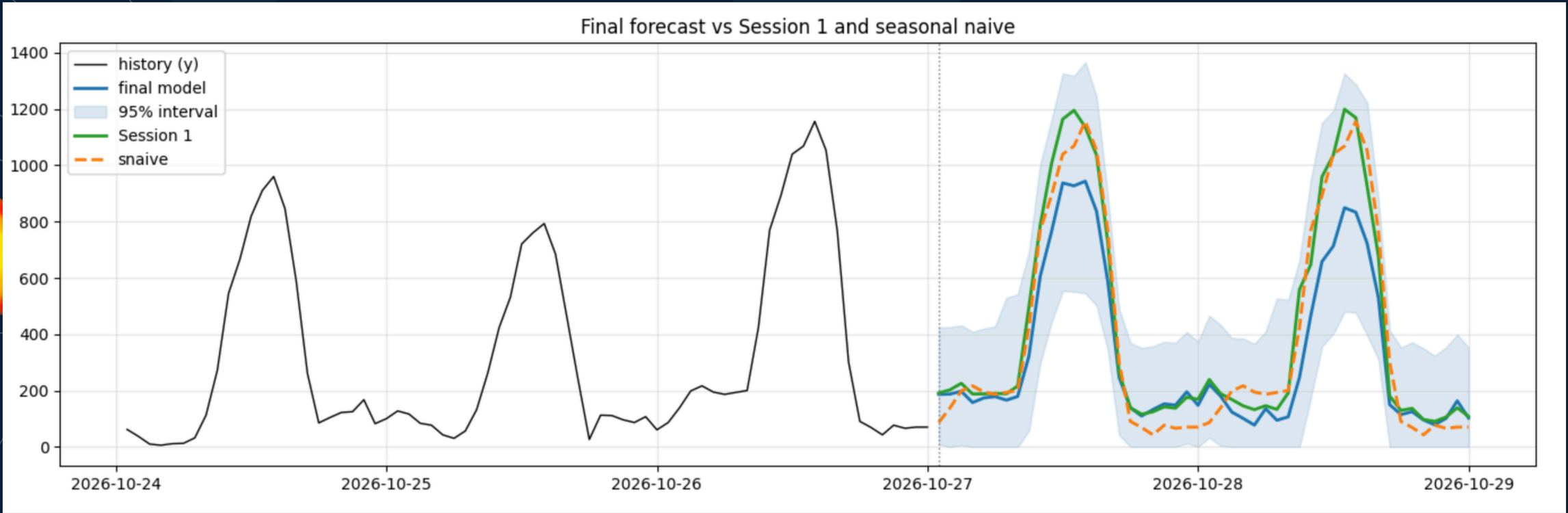
The average of two 95% intervals: quantiles of past residuals and conformalised quantile regression (CQR). Both are calibrated on the last 42 days, separately for day and night, and scale with the current level

VALIDATION

20 windows of 48 hours, one every 4 days from August to October. Each forecast uses only data before its start. Two interval scales are chosen on these windows

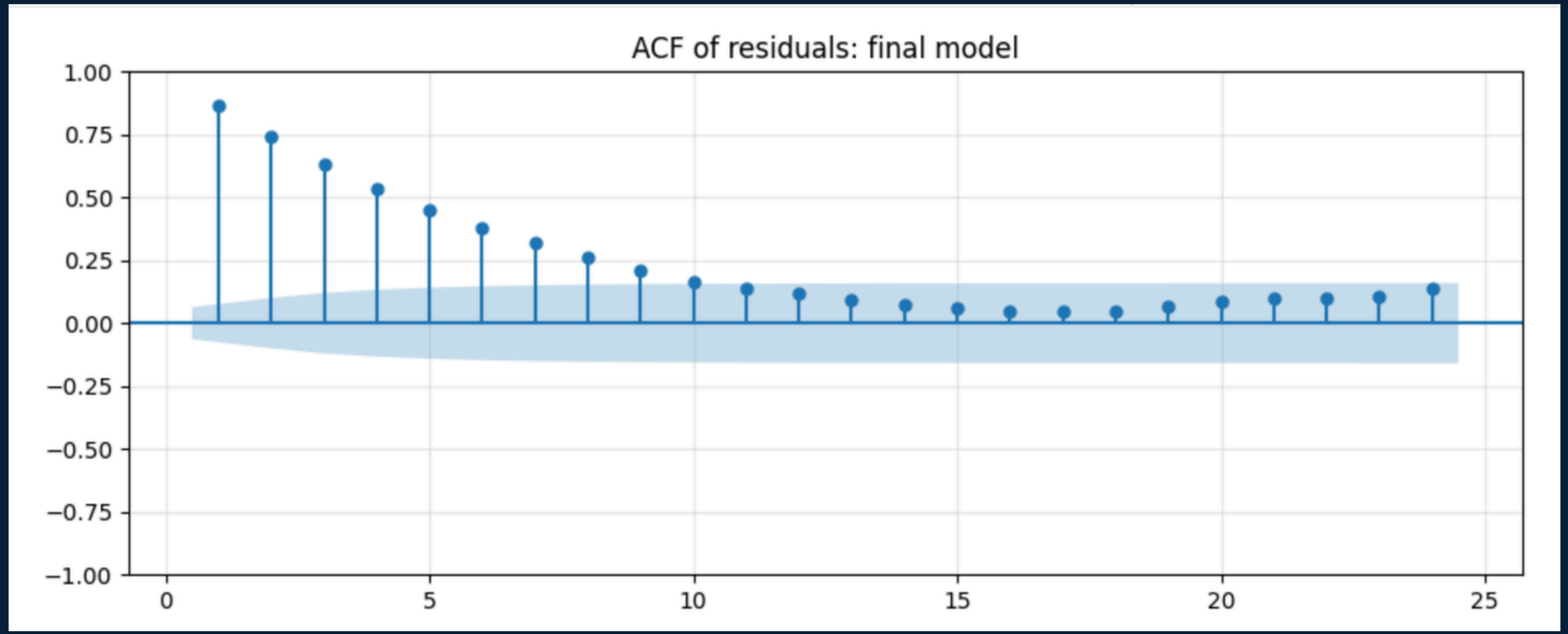
Final model vs snaive on 20 validation windows

- RMSE 208 vs 273
- Winkler 1061 vs 1998
- better than snaive on day 2 in ~70% of windows



What do residuals show?

- Errors persist for hours: 0.87 at lag 1
- Not white noise: Ljung–Box p ≈ 0
- Whole days are missed, not single hours
- Interval holds: 95.3% coverage



What I would do differently

VALIDATE EARLY

Check every model on many windows from the start. The single template window was misleading: seasonal naive RMSE 120 there, 273 on average

EXPLORE THE DATA

The changing level and the role of wind were visible in the data from the start. Seeing them in Session 1 would have shaped the first model

NEXT STEPS

A model for the daily level, a separate model for daytime peaks, and training on noisy weather so the model relies on the forecast less

Thank you for your attention

Now, I'll to answer your questions

